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Inherited Disorders of Sodium and Water Handling

Detlef Bockenhauer

INTRODUCTION

Glomerular filtration yields about 150 L of water and 21,000 mmol of sodium (Na) in a healthy individual in 24 hours, yet only minute fractions of these quantities are excreted eventually in the urine. The precise amount excreted is tightly regulated to maintain homeostasis. A large number of different tubular transporters are involved in this homoeostatic process and either directly or indirectly dependent on the electrochemical gradient generated by the Na^+,K^+-ATPase. In this way, Na reabsorption is central not only for volume but also for acid-base and electrolyte homeostasis. This molecular integration of homeostatic processes is reflected in the clinical picture: whereas disorders of water transport primarily cause a disturbance of serum Na concentration, disorders of Na transport generate clinical signs and symptoms related to blood pressure (BP), electrolytes, and acid-base disturbances, which help establish the diagnosis.

PHYSIOLOGY OF SODIUM AND WATER REABSORPTION

Sodium Transporters and the Corresponding Inherited Disorders

In all tubular epithelial cells, a basolateral energy-requiring Na^+,K^+-ATPase will ensure that intracellular Na is kept at low levels, whereas K is high. Coupled with apical K channels, which use the concentration gradient to establish a lumen-positive electrical potential, an electrochemical gradient for Na is created across the apical cell membrane, which drives Na uptake from the tubular lumen into the cell via dedicated transporters and channels.

Based on anatomic and molecular characteristics, the tubule is typically divided in four functional segments: the proximal tubule (PT), the loop of Henle and for sodium handling specifically the thick ascending limb (TAL), the distal convoluted tubule (DCT), and the collecting duct (CD). Each of these segments has specific apical Na transporters or channels (Fig. 49.1). Transport across the basolateral membrane to complete Na reabsorption is facilitated by the Na^+,K^+-ATPase in all segments. Na transport in these segments is inhibited by specific diuretics, which can help understand inherited disorders of Na transport by comparing them to the effects of the corresponding diuretic.

In the PT, an apical Na-hydrogen (H) exchange protein (NHE3) facilitates most of the Na reabsorption. Because of this connection with H excretion (and thus acid-base homeostasis), this transport process is indirectly blocked by carbonic anhydrase inhibitors, such as acetazolamide. So far, no inherited disorder of isolated Na in the PT has been reported. Instead, impaired reabsorption in the PT typically occurs with a general proximal dysfunction, which underlies the renal Fanconi syndrome (Chapters 13 and 50).

In the TAL, the Na^+-K^+-$2Cl^-$ cotransporter (NKCC2) can be blocked by loop diuretics, such as furosemide and bumetanide. The corresponding inherited disorder is Bartter syndrome, where the function of this transporter is either directly or indirectly impaired. Thus, Bartter syndrome can be conceptualized as "inherited furosemide use."

In the DCT Na is reabsorbed by the Na-Cl cotransporter (NCCT), which is inhibited by thiazides. Disorders involving this transporter include Gitelman syndrome, caused by inherited loss-of-function of this transporter. Gitelman syndrome can therefore be conceptualized as "inherited thiazide use." The mirror image of Gitelman syndrome is pseudohypoaldosteronism type 2 (PHA2) or Gordon syndrome. The underlying genetic causes lead to an overactivity of NCCT.

In the CD, Na reabsorption is facilitated by the epithelial sodium channel (ENaC). Amiloride and triamterene specifically block ENaC. Several inherited disorders affect transport through this channel. Inherited loss-of-function of ENaC is called pseudohypoaldosteronism type 1 and is autosomal recessive, whereas gain-of-function underlies Liddle syndrome. Because ENaC is highly regulated by the mineralocorticoid receptor (MR), several other disorders affect ENaC function through dysregulation of the MR. These and other disorders are discussed in detail later and listed in Table 49.1.

Water Reabsorption

In the PT, water follows sodium chloride (NaCl) passively through aquaporins, the constitutively open water transport proteins in apical and basolateral tubular cell membranes. In this way Na transport is directly linked to unregulated water reabsorption. The TAL and the CD are the 2 segments critical for regulated water reabsorption. The TAL is water impermeable and thus also called the diluting segment, as salt transport via NKCC2 dilutes the tubular fluid. This also helps establish the interstitial concentration gradient, which drives water reabsorption in the arginine-vasopressin (AVP)-sensitive parts of the nephron, the late DCT and CD. In this way, sodium transport is also linked to regulated water reabsorption. Consequently, patients with Bartter syndrome have impaired urinary concentrating ability, which helps explain the polyuria associated with this disorder.

Final control over water reabsorption is exerted in the CD and regulated by AVP. In the absence of this hormone, the apical membrane of the epithelial cells in CD are water impermeable, resulting in water diuresis. Binding of AVP to the AVPR2 receptor in the CD initiates a signaling cascade that results in the insertion of the water channel aquaporin 2 (AQP2) into the apical membrane, allowing water to pass through the epithelial cell layer, following the interstitial concentration gradient, resulting in antidiuresis. Inherited disorders affecting the function of AVPR2 or AQP2 cause either nephrogenic diabetes insipidus (NDI) through loss-of-function or nephrogenic syndrome of inappropriate antidiuresis (NSIAD) through gain-of-function.

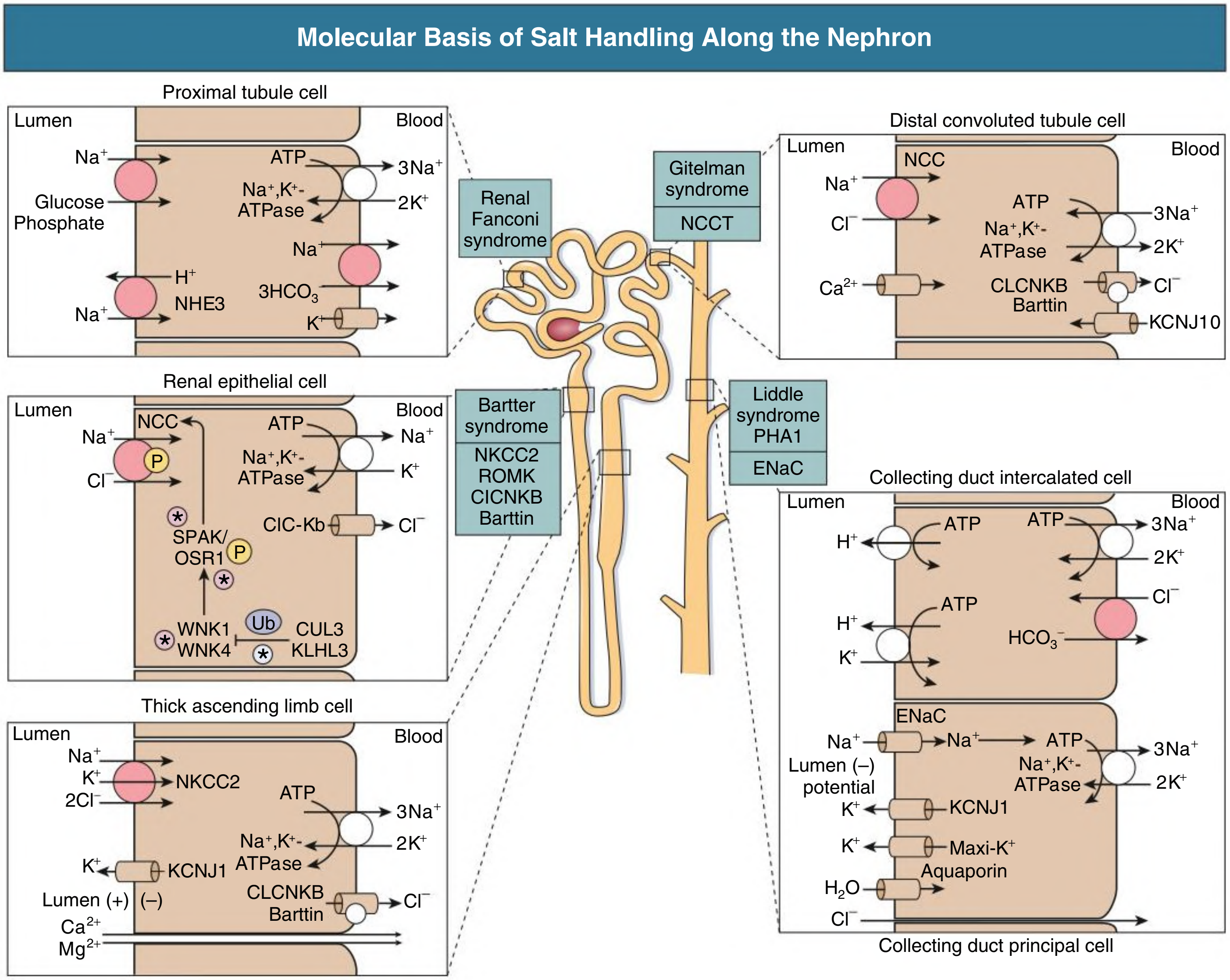

Fig. 49.1 **Overview of the Molecular Basis of Salt Handling Along the Nephron in the Four Main Functional Subunits of the Tubule.** Note that Na^+,K^+-ATPase is expressed on the basolateral aspect of the epithelial cells along the entire nephron, establishing the electrochemical gradient for Na reabsorption and providing an exit pathway for Na. In contrast, on the apical side, segment-specific transporters and channels provide the entry pathway for Na into the epithelial cells. *Salt transport in the proximal tubule:* The majority of filtered salt (roughly 70%–80%) is reabsorbed here through a variety of transporters. The electrochemical gradient for Na is used as driver for a number of transport processes, thereby linking Na reabsorption to acid-base and phosphate homeostasis, as well as glucose and other substrate transport. *Salt transport in the thick ascending limb (TAL) of the loop of Henle:* Approximately 10% to 20% of filtered Na is reabsorbed in this segment with the furosemide-sensitive Na^+-K^+-$2Cl^-$ (NKCC2, variants of which cause Bartter syndrome type 1) cotransporter providing entry into the epithelial cell. Apical K recycling through the KCNJ1 (ROMK) channel (the cause of Bartter syndrome type 2) ensures the efficient functioning of NKCC2 and helps establish a lumen-positive transepithelial voltage that drives paracellular cation reabsorption. Chloride exits the cell through basolateral Cl channels, predominantly CLCNKB (the cause of Bartter syndrome type 3). The β subunit (Barttin, cause of Bartter syndrome type 4) is crucial for normal functioning of CLCNKB. *Salt transport in the distal convoluted tubule (DCT):* Approximately 5% to 10% of filtered Na is reabsorbed in this segment via the thiazide-sensitive Na-Cl-cotransporter (NCCT, the cause of Gitelman syndrome) on the apical side. Note the presence of the same chloride channel (CLCNKB) on the basolateral side as in TAL. This explains why pathogenic variants in this channel cause Bartter syndrome type 3 (a disorder of TAL) but can phenocopy Gitelman syndrome (a disorder of DCT). *Salt transport in a principal cell of the collecting tubule:* Approximately 2% to 5% of filtered Na is reabsorbed in this segment via the amiloride-sensitive epithelial sodium Na channel (ENaC). Loss-of-function variants in this channel cause pseudohypoaldosteronism type 1 (PHA1), whereas gain-of-function causes Liddle syndrome. Na uptake is indirectly coupled to K (through ROMK) and proton secretion through the H^+-ATPase in the neighboring intercalated cell. Aldosterone activates the mineralocorticoid receptor, which increases the activity of ENaC and Na^+,K^+-ATPase, which increases Na reabsorption and K and proton secretion, resulting in hypokalemic alkalosis. Cortisol is also a ligand for the mineralocorticoid receptor but is normally removed by oxidation by 11β-hydroxysteroid dehydrogenase to cortisone. *ATP,* Adenosine triphosphate; *WNK,* with-no-lysine kinase. (From Hoenig MP, Zeidel ML. Homeostasis, the milieu interieur, and the wisdom of the nephron. *Clin J Am Soc Nephrol.* 2014;[9]:1272–1281.)

TABLE 49.1 **Inherited Disorders of Tubular Sodium and Water Handling**

	Inheritance	Gene	Protein	OMIM
Thick Ascending Limb				
Bartter type 1	AR	*SLC12A1*	NKCC2	#600839
Bartter type 2	AR	*KCNJ1*	ROMK	#600359
Bartter type 3	AR	*CLCNKB*	CLC-Kb	#602023
Bartter type 4a	AR	*BSND*	Barttin	#606412
Bartter type 4b	Digenic	*CLCNKA+CLCNKB*	CLC-Ka+CLC-Kb	#602024
Bartter type 5	XR	*MAGED2*	MAGED2	#601199
Autosomal dominant hypocalcemia	AD	*CaSR*	Calcium-sensing receptor	#601198
Distal Convoluted Tubule				
Gitelman syndrome	AR	*SLC12A3*	NCCT	#600968
EAST/SeSAME syndrome	AR	*KCNJ10*	Kir4.1	#602028
PHA type 2b	AD	*WNK4*	WNK4	#601844
PHA type 2c	AD	*WNK1*	WNK1	#605232
PHA type 2d	AD/AR	*KLHL3*	KLHL3	#614495
PHA type 2e	AD	*CUL3*	CUL3	#614496
Collecting Duct				
PHA type 1	AR	SCNN1A	ENaCα subunit	#600228
PHA type 1	AR	SCNN1B	ENaCβ subunit	#600760
PHA type 1	AR	SCNN1G	ENaCγ subunit	#600761
PHA type 1A	AD	NR3C2	MR	#600983
Liddle syndrome	AD	SCNN1B	ENaCβ subunit	#600760
Liddle syndrome	AD	SCNN1G	ENaCγ subunit	#600761
AME	AR	*HSD11B2*	11-β-HSD2	#614232
GRA	AD	*CYP11B1/CYP11B2*	11-β-hydroxylase	#610613
NDI	XLR	*AVPR2*	AVPR2	#300538
NDI	AR/AD	*AQP2*	AQP-2	#107777
NSIAD	XLR	*AVPR2*	V2R	#300538
NSIAD	AD	*GNAS*	G-α s	

AD, Autosomal dominant; *AME,* apparent mineralocorticoid excess; *AR,* autosomal recessive; *AVP,* arginine vasopressin; *GRA,* glucocorticoid remediable aldosteronism; *NDI,* nephrogenic diabetes insipidus; *NSIAD,* nephrogenic syndrome of inappropriate antidiuresis; *PHA,* pseudohypoaldosteronism.

DISORDERS OF SODIUM HANDLING

Based on the clinical and biochemical phenotype, five main categories can be distinguished. Algorithms to help establish the individual diagnoses are shown in Figs. 49.2 and 49.3:

1. Hypokalemic acidosis with low-normal BP (renal Fanconi syndrome and distal renal tubular acidosis, discussed in Chapters 13 and 50)
2. Hypokalemic alkalosis and low-normal BP (Bartter syndrome, Gitelman syndrome, EAST syndrome)
3. Hypokalemic alkalosis and high BP (Liddle syndrome, apparent mineralocorticoid excess [AME], glucocorticoid-remediable hyperaldosteronism [GRA], adrenal 17α-hydroxylase deficiency, adrenal 11β-hydroxylase deficiency, Cushing disease)
4. Hyperkalemic acidosis and low-normal BP (pseudohypoaldosteronism type 1, adrenal 21-hydroxylase deficiency, adrenal aldosterone synthase deficiency)
5. Hyperkalemic acidosis and high BP (Gordon syndrome)

Some of the disorders are caused by mutated renal transport proteins (e.g., Gitelman syndrome) or associated regulatory proteins (e.g., Gordon syndrome). In others, the genetic defect resides in the adrenals, and the changes in adrenal mineralocorticoids and glucocorticoids create the renal phenotype. All the inherited disorders discussed here are rare, with estimated incidences typically around 1:100,000.

CONDITIONS WITH HYPOKALEMIA, METABOLIC ALKALOSIS, AND LOW-NORMAL BP

Bartter Syndrome

Bartter syndrome is a genetically heterogeneous disorder of salt reabsorption in the TAL.[1]

Pathogenesis

Currently, four causative genes have been associated with Bartter syndrome: *SLC12A1* (encoding NKCC2) for Bartter syndrome type 1; *KCNJ1* (encoding KCNJ1, also called ROMK) for type 2; *CLCKNB* (encoding CLCK-B), type 3; and *BSND* (encoding Barttin), type 4 (Table 49.1; see also Fig. 49.1). In addition, another disorder called familial hypocalcemic hypercalciuria, caused by activating mutations in *CaSR,* encoding the calcium-sensing receptor, can cause Bartter-like electrolyte abnormalities and is thus sometimes referred to as Bartter type 5.[2] Although all the genetic defects directly or indirectly impair salt reabsorption through NKCC2, there are differences between the various forms of Bartter syndrome that typically allow correct classification on clinical grounds (see Fig. 49.2). An important first discriminator is the urine calcium excretion. The TAL is an important segment for calcium reabsorption, which occurs passively through paracellular pathways, lined by claudins, which provide transport specificity.[3] The driving force for this is generated by the combined action of NKCC2

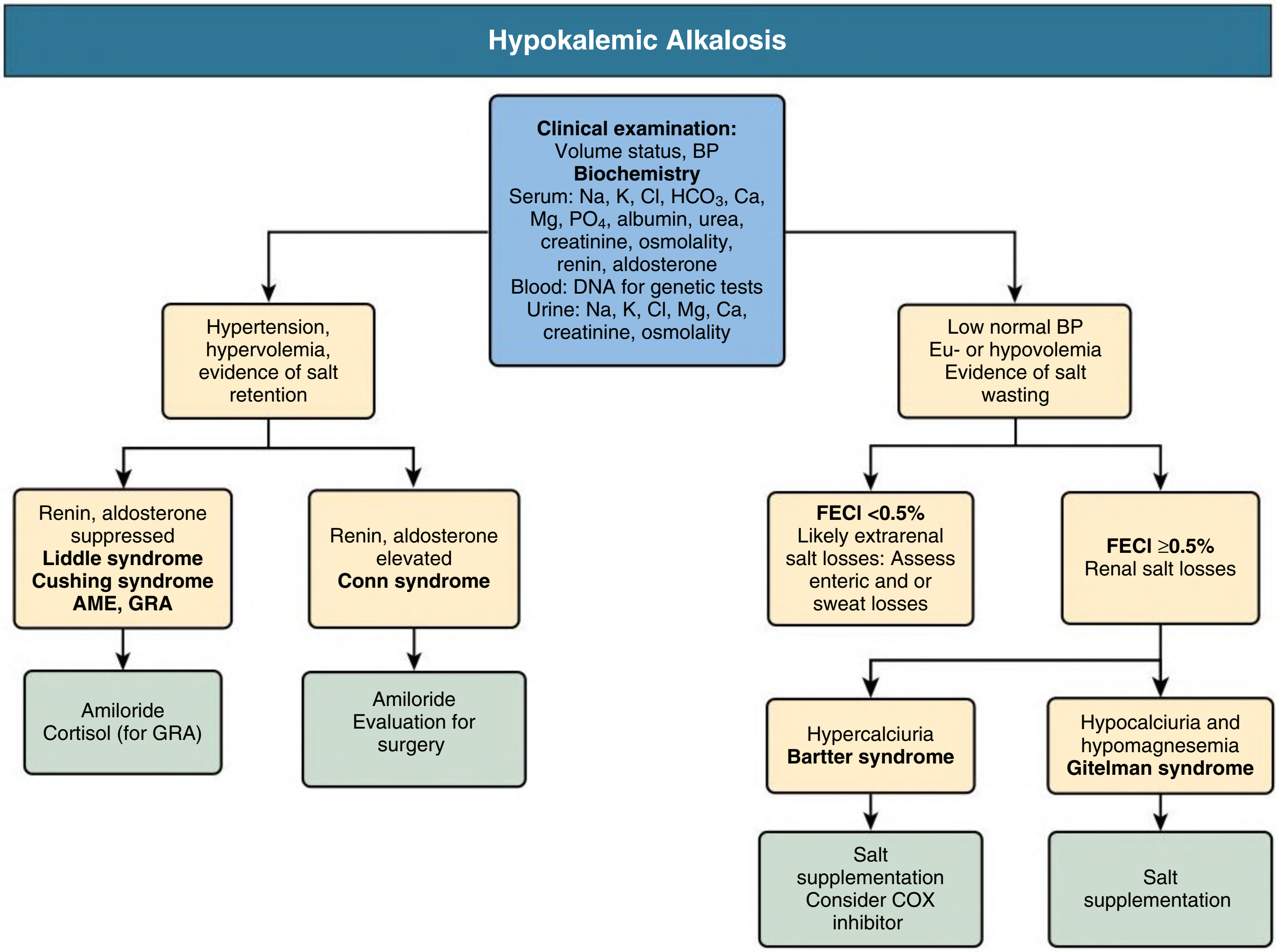

Fig. 49.2 Diagnostic and management algorithm for inherited conditions causing hypokalemic alkalosis. *AME,* Apparent mineralocorticoid excess; *BP,* blood pressure; *COX,* cyclooxygenase; *FECl,* fractional excretion of chloride; *GRA,* glucocorticoid-remediable aldosteronism.

and KCNJ1; transport via NKCC2 is electroneutral, whereas K is recycled back into the tubular lumen via KCNJ1, establishing a lumen positive transepithelial potential. Consequently, calcium reabsorption in the TAL is impaired in Bartter types 1 and 2, leading to hypercalciuria and nephrocalcinosis, whereas it is typically unaffected in types 3 and 4. CLCKNB is also expressed in the DCT, so symptoms of Bartter type 3 can overlap with Gitelman syndrome. Hypomagnesemia is thus a common feature. Finally, Barttin is critical also for inner ear function, so affected patients not only have Bartter syndrome but also sensorineural deafness.

The macula densa is part of the TAL, and it is in the macula densa that the first initial step of tubuloglomerular feedback (TGF) is localized: reduced Cl absorption in the macula densa initiates a signaling cascade that includes enhanced prostaglandin E2 (PGE2) production with consequent activation of renin and aldosterone.[4] Because Cl absorption is impaired in Bartter syndrome, these patients have impaired TGF with elevated levels of PGE2, renin, and aldosterone, and the latter mediates the characteristic hypokalemic metabolic alkalosis.[5] The dramatic upregulation of TGF is reflected in hypertrophy of the juxtaglomerular apparatus.[6]

Clinical Manifestations

Bartter syndrome from pathogenic variants in *SLC12A1, KCNJ1,* or *BSND* usually has earlier onset than that caused by variants in *CLCKNB.* Most manifest already during pregnancy, leading to the term *antenatal* Bartter syndrome.[7,8] In contrast, the later onset form is often called *classic* Bartter syndrome, as this is the phenotype originally described by Bartter and colleagues.[6]

The clinical features of antenatal Bartter syndrome include polyuria, hypokalemic metabolic alkalosis, and high urinary chloride excretion in a newborn with vomiting, failure to thrive, and a history of polyhydramnios and premature delivery. In pregnancies affected by polyhydramnios, a prenatal diagnosis of Bartter syndrome may be suggested by elevated protein concentrations in amniotic fluid, if genetic testing is not informative.[9] Bartter syndrome type 2 (BS2) can have a different phenotype in the first days of life, as KCNJ1 is not only involved in salt reabsorption in the TAL but also mediates K secretion in CD. Consequently, patients with BS2 often present initially with hyperkalemia and hyponatremia, leading to an erroneous diagnosis of pseudohypoaldosteronism type 1 (PHA1).[10] Over the first few days to weeks, serum K levels decrease, presumably due

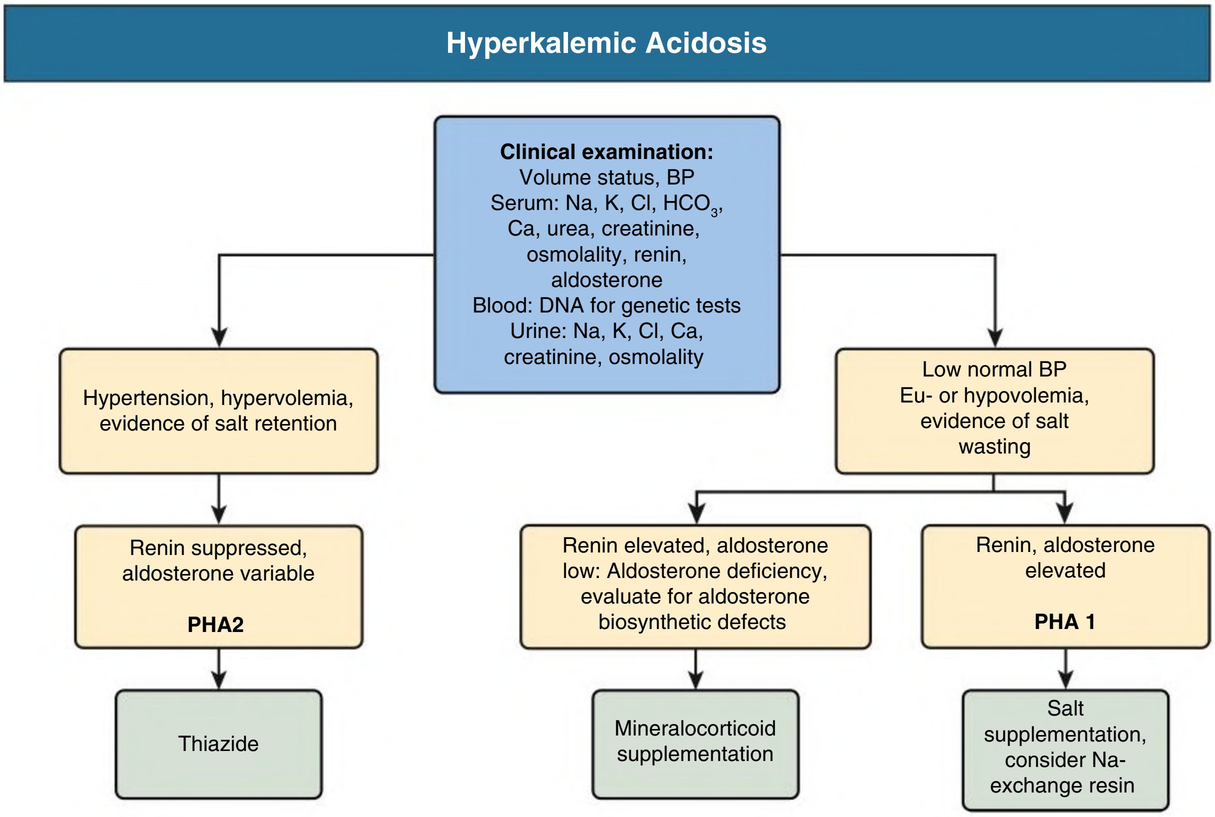

Fig. 49.3 Diagnostic and management algorithm for inherited conditions causing hyperkalemic acidosis. *BP*, Blood pressure; *PHA1*, pseudohypoaldosteronism type 1; *PHA2*, pseudohypoaldosteronism type 2.

to the expression of other K channels in CD that can compensate for the loss of KCNJ1 function.

Hypercalciuria and nephrocalcinosis are typical for BS1 and BS2, as the coupling of the electroneutral NKCC2 transporter with recycling of potassium into the tubular lumen through KCNJ1 is establishing the transepithelial voltage that drives Ca (and Mg) reabsorption through the paracellular pathway.[3]

Classic Bartter syndrome is typically associated with pathogenic variants in *CLCNKB* (Bartter syndrome type 3 or BS3). These patients present mostly in the first decade of life with vomiting, polyuria, recurrent episodes of dehydration, and hypokalemic metabolic alkalosis. Interestingly, electrolyte abnormalities are typically more severe than in antenatal Bartter syndrome.[7,8] Moreover, hypomagnesemia with renal Mg^{2+} wasting is common. Presumably, this is due to the expression of CLC-Kb in both TAL and DCT, thus affecting salt reabsorption in both these nephron segments. This may also explain the phenotypic overlap between Bartter and Gitelman syndrome, as some patients with BS3 can phenotypically mimic Gitelman syndrome.[11]

Bartter syndrome type 4 (BS4) is also characterized by sensorineural deafness. Clinically, it typically manifests as a more severe form of BS3, sometimes with extreme electrolyte abnormalities.[5] Presumably, in BS3, the closely related chloride channel CLC-Ka can compensate for the loss of CLC-Kb function in the inner ear and to some degree also in the kidney. However, both channels require Barttin as subunit, so that loss of function of Barttin is functionally equivalent to loss of both chloride channels.[12] Of note, BS4 patients typically experience progressive chronic kidney disease (CKD), although a milder phenotype has been reported, presumably to mutations with residual Barttin function.[13]

Diagnosis

The key features of Bartter syndrome are hypokalemic, hypochloremic alkalosis with evidence of renal Cl and K wasting and low or normal BP.

Differential Diagnosis

In syndromes of chronic severe hypokalemia with metabolic alkalosis, the differential diagnosis may be facilitated greatly by taking into consideration the associated BP and the urinary chloride concentration (Fig. 49.4).

Hypertension with hypokalemic alkalosis indicates a primary activation of salt reabsorption in the CD, such as from pathologic activation of the MR (e.g., hyperaldosteronism) or the sodium channel ENaC. If Bartter syndrome is associated with a normal or low-normal BP, however, extrarenal or renal Cl and Na losses are the cause. Extrarenal loss of sodium occurs in diarrhea, vomiting, or burns and is characterized by very low Cl concentrations in the urine with a fractional chloride excretion less than 0.5%.[14] Congenital chloride diarrhea can be misdiagnosed as BS, as the biochemical picture in the blood is identical, and the large and often clear-appearing stool volume is misinterpreted as urine, especially in female babies in diapers.[15] It is critical to demonstrate the renal loss of salt that is typical of Bartter and Gitelman syndrome, and diuretic use. Further analysis of Mg and urinary Ca excretion can then help distinguish between the various forms of Bartter and Gitelman syndrome (Table 49.2). Genotyping is recommended to confirm the final diagnosis.

Treatment

Neonates with Bartter syndrome typically have marked fluid and electrolyte disturbances that need to be corrected carefully. Salt

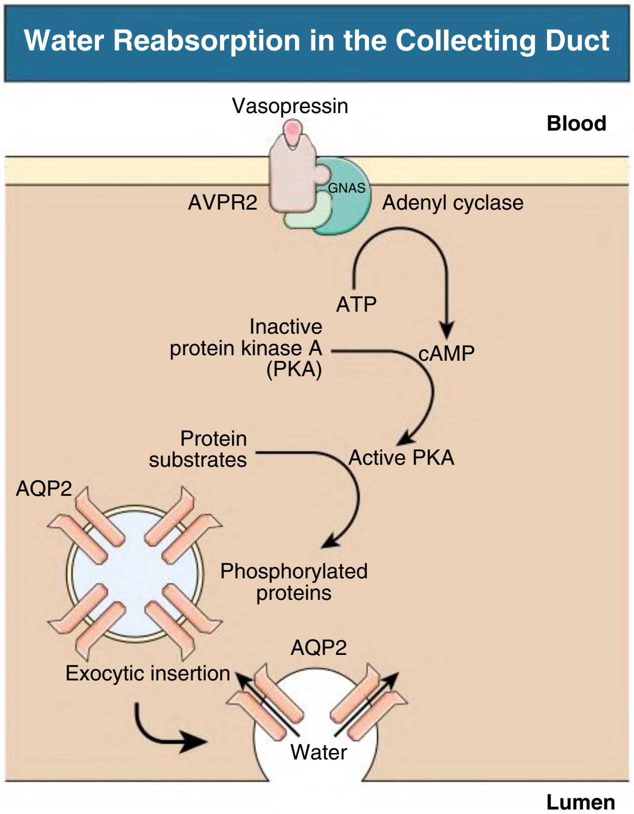

Fig. 49.4 Activation of the basolateral arginine-vasopressin type 2 receptor (AVPR2) initiates the signaling cascade that ultimately results in insertion of the water channel aquaporin 2 (AQP2) in the apical membrane, making it water permeable and thus allowing water reabsorption and urinary concentration. Genetic variants leading to loss-of-function of AVPR2 cause X-linked nephrogenic diabetes insipidus, whereas those with gain-of-function cause nephrogenic syndrome of inappropriate antidiuresis (NSIAD). Similarly, gain-of-function variants in the stimulatory G-protein alpha subunit also cause NSIAD. Loss-of-function variants in AQP2 are the cause of autosomal nephrogenic diabetes insipidus. *ATP,* Adenosine triphosphate; *cAMP,* 3′,5′-cyclic adenosine monophosphate; *GNAS,* genome-wide association studies; *PKA,* protein kinase A.

supplementation is often helpful; however, antenatal Bartter syndrome can be complicated by a secondary form of NDI, in which case a high solute intake can lead to hypernatremia.[16] Potassium chloride supplementation is commonly provided. Addition of spironolactone or amiloride may be useful in improving hypokalemia and alkalosis but worsens the salt wasting and thus increases the risk of hypovolemia and should only be provided with sufficient salt supplementation.[5] The same applies for angiotensin-converting enzyme (ACE) inhibitors, which are not recommended.[17]

A key problem in renal wasting disorders is that supplementation primarily results in increased urinary losses. After supplementation, plasma levels transiently rise, leading to an increased concentration in the glomerular filtrate with consequent loss in the urine and decreasing plasma levels. Monitoring the efficacy of supplementation thus depends strongly on the timing of the blood sample. Although cardiac arrhythmias and other severe complications have been reported with marked hypokalemia,[18] the dramatic swings in plasma concentration associated with high-dose supplementation may be more harmful than consistent levels below the normal range. Moreover, high-dose supplementation may cause gastrointestinal symptoms (e.g., ulcers, diarrhea) that can worsen the electrolyte profile. Thus, smaller but more frequent administration of electrolyte supplements may be better tolerated and safer.

A key component of treatment especially of antenatal BS is prostaglandin synthesis inhibition through cyclooxygenase inhibition by nonsteroidal antiinflammatory drugs (NSAIDs) such as indomethacin (1–3 mg/kg/24 h). Selective COX2 inhibitors may provide similar efficacy with less potential toxicity.[19] NSAIDs act by interfering with the pathologic tubuloglomerular feedback in BS and enhancing proximal sodium reabsorption.[5] Treatment results in reduction of polyuria and polydipsia, improved growth and activity, and better control of plasma electrolytes. Plasma levels of renin and aldosterone often decrease, sometimes into the normal range.

Outcomes

Dramatic complications such as intracranial hemorrhage and bronchopulmonary dysplasia are typically seen in patients with antenatal BS and are a consequence of extreme prematurity. As with most diseases, there is a spectrum of severity. The electrolyte and acid-base imbalance itself can be severe with complications such as arrhythmias, paralysis, rhabdomyolysis, and apnea, yet others may have few symptoms, and some patients are diagnosed incidentally later in life. Importantly, even patients with more severe electrolyte imbalance usually improve clinically with appropriate therapy. Progressive CKD can be a complication of prematurity but can also be seen in BS3 patients born at term and is common in BS4.

Gitelman Syndrome

Gitelman syndrome is an autosomal recessive condition characterized by hypokalemic hypochloremic metabolic alkalosis with hypocalciuria and hypomagnesemia. The prevalence of Gitelman syndrome is estimated at approximately 1:40,000 in White populations, making it arguably the most frequent inherited tubulopathy.[20]

Pathogenesis

The similarity between the features of Gitelman syndrome and those caused by thiazide administration originally suggested that the defect might be in the DCT, which was subsequently confirmed by genetic studies that linked the condition to inactivating mutations in *SLC12A3*, the gene encoding the thiazide-sensitive transporter NCCT (see Fig. 49.1 and Table 49.1).[21] Loss of NCCT function results in Na and Cl wasting from this segment, leading to hypovolemia with secondary activation of the renin-aldosterone system. As in Bartter syndrome, the resulting increase in CT Na reabsorption is counterbalanced by K and H excretion, causing hypokalemic alkalosis.

The hypocalciuria may be caused by enhanced proximal tubular calcium reabsorption, secondary to plasma volume contraction.[22] The renal magnesium wasting is caused by downregulation of the epithelial magnesium channel TRPM6 in DCTs.

Clinical Manifestations and Diagnosis

The severity of symptoms varies widely. More severely affected patients complain of a decreased quality of life with fatigue and muscle weakness, inability to work for extended periods, salt craving, and polyuria.[23] Cardiac disturbances, muscle cramps, and tetany are only exceptionally present. Chondrocalcinosis and sclerochoroidal calcifications can occur later in life.[20] Laboratory evaluation shows hypokalemic hypochloremic metabolic alkalosis with hypocalciuria and hypomagnesemia with elevated urinary Cl, K, and Mg excretion. BP will usually be in the low-normal range. The differential diagnosis includes Bartter syndrome, especially BS3, HNF1B-related disease, and EAST syndrome, as well as acquired disorders, such as thiazide abuse and Sjögren syndrome.[20] Genotyping should be performed to confirm the diagnosis.

TABLE 49.2 **Distinguishing Features of Individual Bartter and Gitelman Syndromes**

	BARTTER					
Feature/Syndrome	**Type 1**	**Type 2**	**Type 3[a]**	**Type 4**	**Type 5[b]**	**Gitelman**
Antenatal manifestations (polyhydramnios)	Common	Common	Variable	Common	Always	Absent
Age at presentation	Neonate	Neonate	Variable	Neonate	Neonate	Childhood/adolescence
Polyuria, polydipsia	Marked	Marked	Rare	Common	Marked	Rare
Neonatal hyperkalemia/acidosis	Absent	Common	Absent	Absent	Absent	Absent
Spontaneous resolution	Absent	Absent	Absent	Absent	Always	Absent
Urinary calcium excretion	Elevated	Elevated	Variable	Normal	Elevated	Low
Nephrocalcinosis	Common	Common	Rare	Rare	Rare	Absent
Hypomagnesemia	Rare	Rare	Common	Common	Rare	Common
Urine prostaglandins	Elevated	Elevated	Variable	Elevated	Elevated	Normal

[a]Note the intermediate phenotype of Bartter syndrome type 3, which can present with typical features of Bartter syndrome but also phenocopy Gitelman syndrome and even switch between these two phenotypes in an individual.
[b]As Bartter syndrome type 5 spontaneously resolves typically around the estimated date of delivery, features detailed here only are present during the active disease period.

Treatment

Gitelman syndrome is primarily a salt-wasting disorder, and treatment should therefore include a liberal salt intake, if not pharmacologic supplementation. K and Mg supplements are usually given to improve muscle weakness or cramps. However, dosing may be limited by diarrhea and abdominal discomfort, and values in the normal range are typically not achieved. In exceptional cases of severe symptoms, parenteral Mg has been infused. NSAID are usually not helpful. K-sparing diuretics may improve the biochemistry but compound the salt wasting and should thus be considered carefully.

The long-term prognosis is good for cardiac and renal function, as well as for general health, although some patients do complain about markedly impaired quality of life. There is no clear association between severity of the electrolyte imbalance and symptoms, which complicates adjustment of treatment.

CONDITIONS WITH HYPOKALEMIA, METABOLIC ALKALOSIS, AND HYPERTENSION

Conditions with hypokalemia, metabolic alkalosis, and hypertension are caused by primary activation of the MR or its downstream effector, the epithelial Na-channel ENaC (see Fig. 49.1).

Liddle Syndrome

Liddle syndrome is an autosomal dominant disorder caused by gain-of-function of ENaC.[24] Consequently, the enhanced Na reabsorption through ENaC is independent of MR activation. Renin and aldosterone levels are suppressed, and there is no response to MR blockers, such as spironolactone or eplerenone. However, triamterene and amiloride, which are blockers of ENaC, correct hypertension, renal K loss, and alkalosis.[25]

Pathogenesis

Liddle syndrome is caused by variants in genes encoding the β (*SCNN1B*) or γ (*SCNN1G*) subunit of ENaC. The majority of patients carry a variant that impairs retrieval of the channel from the membrane, leading to enhanced ENaC activity.[26] The variants result in truncations of the cytoplasmic C-terminal tail of the affected subunits. CD Na reabsorption depends on the channel density present in the apical cell membrane. Channel density is regulated by removal of ENaC from the cell membrane by ubiquitination and degradation. In Liddle syndrome, the mutated ENaC protein cannot be recognized by NEDD4, a ubiquitin ligase protein; thus, the channels remain in the cell membrane for prolonged periods. This action results in enhanced sodium reabsorption, hypertension, and hypokalemic alkalosis (see Fig. 49.1).

Clinical Manifestations and Diagnosis

Liddle syndrome is a rare disorder of hypertension that typically first presents in teenage children with hypokalemic metabolic alkalosis and low blood levels of renin and aldosterone. However, clinical manifestation as early as the newborn period has been described.

This condition is easily distinguished from primary hyperaldosteronism or renal artery stenosis by the finding of low renin and aldosterone levels. Other conditions with similar phenotype include AME and GRA (Table 49.3), as well as 11β-hydroxylase (steroid 11β-monooxygenase) or 17α-hydroxylase (steroid 17α-monooxygenase) deficiency and can be separated by urinary steroid profiles and genetic testing. An activating mutation of the MR with exacerbation of hypertension in pregnancy has also been reported and should be differentiated.[27]

Treatment

Therapy consists of ENaC blockade with triamterene or amiloride, which usually normalizes the BP and K levels. Because the pathogenetic disorder is not correctable with age, lifelong therapy is required. Patients who have developed end-stage kidney disease from uncontrolled hypertension do not experience recurrence of the disease after transplantation, as the graft does not contain the genetic alteration.

Apparent Mineralocorticoid Excess

Pathogenesis

AME is an autosomal recessive condition resulting from deficiency of the type II (renal and placental) isoform of the enzyme 11β-hydroxysteroid dehydrogenase. Clinical features of AME are similar to those of Liddle syndrome, but symptoms typically present earlier (in infancy) and more severely.

Normally, aldosterone is the principal ligand for the MR and in this way increases synthesis of various proteins, chiefly Na^+,K^+-ATPase on the basolateral surface and ENaC on the apical surface. These proteins facilitate the increased Na reabsorption and K secretion in the aldosterone-sensitive distal nephron (see Fig. 49.1). Interestingly, MR

TABLE 49.3 Features of Liddle Syndrome, Apparent Mineralocorticoid Excess (AME), and Glucocorticoid-Remediable Aldosteronism (GRA)

Feature/Syndrome	Liddle	AME	GRA
Inheritance	AD	AR	AD
Age at presentation	Childhood/adolescence	Neonate/infancy	Childhood/adolescence
Plasma aldosterone	Suppressed	Suppressed	Elevated
Plasma renin activity	Suppressed	Suppressed	Suppressed
Urinary mineralocorticoid metabolites	Normal	Elevated ratios of THF + allo-THF to THE; free cortisol to cortisone	Elevated cortisol C-18 oxidation products
Response of the Hypertension to:			
Glucocorticoids	No	Yes[a]	Yes
Amiloride	Yes	Yes	Yes
Spironolactone	No	Yes	Yes

These syndromes are all characterized by hypokalemia, metabolic alkalosis, and hypertension.
Response to amiloride is representative for all blockers of the epithelium sodium channel (e.g., triamterene), and response to spironolactone is representative for all blockers of the mineralocorticoid receptor (e.g., eplerenone).
[a]Glucocorticoids are not standard treatment of AME. However, exogenous administration of a glucocorticoid that does not activate the mineralocorticoid receptor (e.g., cortisone) but that suppresses endogenous cortisol production would ameliorate the hypertension.
AD, Autosomal dominant; *AR,* autosomal recessive; *THE,* tetrahydrocortisone; *THF,* tetrahydrocortisol.

is not specific for aldosterone, but can also bind cortisol, which circulates in blood at roughly 1000-fold higher concentration than aldosterone. To provide specificity for aldosterone, the MR is protected by the activity of the 11β-hydroxysteroid dehydrogenase type 2 (HSD11B2), which is highly expressed in the distal nephron and metabolizes cortisol to cortisone, which lacks MR binding ability.

Loss-of-function mutations in HSD11B2 thus cause AME by nonspecific activation of MR by cortisol.[28] Interestingly, carbenoxolone and glycyrrhizic acid (found in licorice compounds) are potent inhibitors of this enzyme and so can cause an acquired form of AME. HSD11B2 is also expressed in the placenta, likely explaining the intrauterine growth retardation typically seen in AME patients.

Clinical Manifestations and Diagnosis

AME is characterized by early onset of severe hypertension in childhood, hypokalemia, metabolic alkalosis with suppressed plasma levels of renin and aldosterone, and increased metabolites of cortisol in the urine. A history of low birth weight and subsequent failure to thrive is typical.[29] Untreated, stroke and other complications of severe hypertension are seen even during childhood. Milder clinical manifestations can be seen with mutations that retain some functionality of the enzyme. Polyuria due to a urinary concentrating defect is another typical feature and likely linked to the electrolyte abnormalities in blood and urine, as it improves after treatment of AME.[30]

Establishing the diagnosis of AME is critical for proper treatment and should be prompted by the electrolyte profile and early-onset hypertension with suppressed renin and aldosterone levels. A urinary steroid profile is diagnostic by showing elevated urinary levels of hydrogenated metabolites of cortisol (tetrahydrocortisol plus allotetrahydrocortisol) compared with cortisone (tetrahydrocortisone). The ratio of urinary-free cortisol to cortisone is also increased. Genetic testing confirms the diagnosis.

Treatment

Treatment with blockers of MR (e.g., spironolactone or eplerenone) or its main effector ENaC (e.g., amiloride or triamterene) provides specific treatment. However, as MR blockers are competitive antagonists and in AME compete against the 1000-fold higher cortisol concentrations (compared with aldosterone), large doses are needed, and ENaC blockers (or a combination of both) are likely to be more effective.

Glucocorticoid-Remediable Aldosteronism

GRA is an autosomal dominant condition. Patients present with typical features of primary hyperaldosteronism: hypertension, suppressed plasma renin activity, and hypokalemia. Unlike primary hyperaldosteronism (due to aldosterone-producing adrenal adenoma), hypersecretion of aldosterone in GRA can be reversed by the administration of corticosteroids. Affected individuals have early-onset hypertension. There is a high prevalence of hemorrhagic stroke, largely as a result of ruptured intracranial aneurysms.

Pathogenesis

Patients with GRA have adrenocorticotropic hormone (ACTH)-sensitive aldosterone production occurring in the zona fasciculata of the adrenal gland, which is usually responsible only for cortisol synthesis. Normally, aldosterone is synthesized in the zona glomerulosa. The two isoenzymes of 11β-hydroxylase involved in the biosynthesis of aldosterone and cortisol are steroid 11β-hydroxylase (CYP11B1) and aldosterone synthase (CYP11B2), respectively. The genes for these isoenzymes are located close to each other on the long arm of chromosome 8. Unequal meiotic crossovers may produce hybrid genes by fusion of the promoter end of *CYP11B1* with the coding sequence of *CYP11B2,* so that aldosterone synthase is inappropriately regulated by ACTH.[31]

Diagnosis

Patients with GRA are often misdiagnosed with primary hypertension. Hypertensive patients with early-onset hypertension, early cerebral hemorrhage (<40 years), hypokalemia before or after diuretic therapy, and refractoriness to standard antihypertensive medication are candidates for GRA testing, especially in the presence of a family history of hypertension or early death. Similar to other genetic forms of hypertension (Liddle syndrome, AME, Gordon syndrome), plasma renin activity is suppressed. Aldosterone levels are normal to high and do not change with posture but are suppressed by dexamethasone.

A urinary steroid profile will show the biochemical hallmark of GRA in the form of increased levels of the so-called hybrid steroids (18-hydroxycortisol and 18-oxocortisol).

Genetic testing confirms the diagnosis.

Treatment

Treatment with low-dose corticosteroid is effective in GRA patients, as it suppresses ACTH and thus the aberrant ACTH-mediated aldosterone production in the zona fasciculata. Typically, 0.125 to 0.25 mg of dexamethasone or 2.5 to 5 mg of prednisolone is administered at bedtime. Therapeutic goals are normotension and normalization of biochemical markers (urinary 18-oxosteroid, serum aldosterone). MR antagonists (spironolactone, eplerenone) and ENaC antagonists such as amiloride and triamterene are also useful treatments.

Adrenal Enzymatic Disorders

Inherited deficiency of 11β- or 17α-hydroxylase, as well as specific heterozygous variants in the ion channels KCNJ5 and CACNA1H, also cause mineralocorticoid excess with hypertension and hypokalemic metabolic alkalosis. These disorders are covered in endocrinological textbooks and thus will not be discussed here in detail. Hypertension is caused by excessive mineralocorticoid production. Again, a urinary steroid profile helps establish the diagnosis, which can be confirmed by genetic testing.

CONDITIONS WITH HYPONATREMIA, HYPERKALEMIA, METABOLIC ACIDOSIS, AND NORMAL BLOOD PRESSURE

Conditions with hyponatremia, hyperkalemia, metabolic acidosis, and low-normal BP have features of mineralocorticoid deficiency either because of a synthetic defect or because of end organ resistance.

Pseudohypoaldosteronism

Pseudohypoaldosteronism (PHA) is a state of renal tubular unresponsiveness to the action of aldosterone.

Type 1 PHA exists in an autosomal dominant (adPHA1) and an autosomal recessive form (arPHA1). The adPHA1 is due to loss-of-function variants in *NR3C2* encoding the MR.[32] In contrast, arPHA1 is due to loss-of-function variants in the genes encoding the α, β, or γ subunits of ENaC.[33]

Diagnosis

PHA1 presents typically in the first few days (arPHA1) to weeks (adPHA1) of life with weight loss, hypovolemia, and poor feeding. Biochemistries will reveal the typical electrolyte profile with mild to moderate hyponatremia, often severe (arPHA1) hyperkalemia and metabolic acidosis. Urine biochemistries reveal inappropriately high Na excretion and virtually absent K secretion. Occasionally, patients with BS2 can present similarly (see "Bartter Syndrome" section). PHA1 can be distinguished from aldosterone deficiency states, such as congenital adrenal hyperplasia, by the massively elevated aldosterone levels in blood. The diagnosis can be confirmed by genetic testing. An acquired form of PHA1 can be seen with urinary tract obstruction and/or pyelonephritis.

Clinical Features

The clinical manifestations are dominated by electrolyte abnormalities and hypovolemia, which untreated can lead to circulatory shock. ENaC is also expressed in the skin and lungs, where it mediates salt reabsorption. Consequently, arPHA1 patients have increased sweat Na concentration and can develop a miliary rash from blockage of the sweat glands. In addition, patients can have cystic fibrosis–like lung disease due to viscous high-salt–containing bronchial secretions.

Treatment

The initial emergency treatment focuses on volume resuscitation with 0.9% saline. Without treatment, PHA1, especially arPHA1, can be lethal. Maintenance treatment consists of NaCl and $NaHCO_3$ supplementation adjusted to maintain euvolemia, normonatremia, and acid-base homeostasis. In arPHA1, Na-exchange resins can help stabilize patients by providing a steady source of Na and an alternative means for K excretion. Patients with adPHA1 typically improve spontaneously over the first few months of life and maintain normal plasma electrolytes without supplementation, although they typically maintain increased renin-aldosterone levels.[34]

Aldosterone Biosynthetic Defects

Patients with defects in aldosterone biosynthesis show salt wasting with hyponatremia, hyperkalemia, hypovolemia, and elevated plasma renin activity yet obviously low or absent aldosterone levels. These conditions are not detailed here.

CONDITIONS WITH HYPERKALEMIA, METABOLIC ACIDOSIS, AND HYPERTENSION

Pseudohypoaldosteronism Type 2 (Gordon Syndrome)

The clinical mirror image of Gitelman syndrome, Gordon syndrome is an autosomal dominant condition characterized by hypertension, hyperkalemia, and mild hyperchloremic metabolic acidosis.

Pathogenesis

Initially, two genes were identified as responsible for PHA2 (Table 49.1).[35] These genes encode two members of the with-no-lysine kinase family (WNK1 and WNK4), both expressed in the convoluted tubule and the CDs. WNK4 acts as a negative regulator of thiazide-sensitive NCCT function, reducing cell surface expression of NCCT. WNK4 also downregulates KCNJ1 and epithelial chloride flux. Pathogenic variants in *WNK4* are typically missense and cause loss-of-function, so that WNK4 loses its ability to suppress NCCT and KCNJ1, leading to Na and K retention. WNK1 prevents WNK4 from interacting with NCCT. Pathogenic variants in *WNK1* are typically intronic deletions that increase WNK1 expression. Subsequently, two further genes were identified variants in which cause PHA2: *CUL3* and *KLHL3*.[36] The encoded proteins form a ubiquitin-ligase complex that regulates WNK1 and WNK4 abundance. Interestingly, in addition to the autosomal dominant inheritance typical for PH2, variants in *KLHL3* can also be recessive. Moreover, pathogenic variants in *CUL3* are frequently de novo, so that the absence of a family history does not exclude the diagnosis. An acquired form of PHA2 can be seen as a side effect of calcineurin inhibitors, especially tacrolimus, which affect WNK activity.[37]

Clinical Manifestations and Diagnosis

Hyperkalemia may be present from birth, but as in GRA, hypertension may not manifest until later in life. Patients show hyperchloremic metabolic acidosis; plasma renin and aldosterone are variable, but renin activity is usually suppressed. Patients with CUL3-related disease appear to be phenotypically more seriously affected with higher plasma K levels, more pronounced acidosis and hypertension and consequently younger age at diagnosis.[36] In addition, most patients with CUL3 disease have failure to thrive and growth impairment.

Treatment

As specific inhibitors of NCCT, thiazides are able to completely correct the clinical and biochemical features of PHA2, except for the extrarenal manifestations in CUL3-related disease. Establishing the diagnosis is thus critical for targeted and effective treatment.

Congenital Nephrogenic Diabetes Insipidus

Congenital NDI is a rare polyuric disorder characterized by the failure to concentrate urine despite elevated levels of vasopressin. Congenital NDI is caused by mutations in key proteins controlling water reabsorption in the CD. Acquired diabetes insipidus (DI) is much more common, and its diagnosis and management are discussed in Chapter 9.

Pathogenesis

More than 90% of patients have X-linked recessive NDI with pathogenic variants in *AVPR2,* the gene for the arginine-vasopressin type 2 receptor (AVPR2). Consequently, the majority of patients are male, although female carriers can have a urinary concentrating defect of variable severity, presumably due to nonrandom X inactivation.[38] AVPR2 is expressed predominantly in the kidney, where it mediates urinary concentration, but to some degree also in the vasculature, where it mediates vasodilation and release of von Willebrand factor.[39] Well over 250 pathogenic variants have been reported so far.[38] In fewer than 10% of the families, congenital NDI has an autosomal recessive inheritance with pathogenic variants in *AQP2,* the gene encoding the water channel AQP2, which constitutes the final effector protein in the vasopressin-initiated signaling cascade. A rare autosomal dominant form of NDI, due to dominant negative variants in *AQP2,* has also been reported. Some inherited diseases, such as Bartter syndrome and AME, can also be associated with so-called secondary inherited NDI.[16] The etiology of the AQP2 deficiency in these disorders is unclear but likely related to the blood and/or urinary electrolyte abnormalities, as both hypokalemia and hypercalciuria have been associated with decreased AQP2 expression.

Clinical Features

Manifestations of congenital NDI typically appear within the first weeks to months of life. NDI does not cause polyhydramnios, as the osmotic load is cleared by the placenta. Clinical features at presentation are typically polyuria and failure to thrive, as well as a history of vigorous sucking followed by vomiting.[38] Unrecognized and untreated, the baby can experience hypernatremic dehydration, with delayed development and mental impairment as possible consequences. Cranial computed tomography may occasionally show dystrophic calcification in the basal ganglia and the cerebral cortex in such untreated patients.

Attention-deficit/hyperactivity disorder and impaired school performance are commonly reported and may be secondary to preoccupation with drinking and voiding. An adult patient usually drinks and voids about 10 to 12 L per day. Urinary tract dilation can be seen, especially if there are voiding abnormalities.

Some variants in either *AVPR2* or *AQP2* may retain partial functionality of the protein, leading to a phenotype of partial NDI, where urinary concentration is possible but subnormal. Clinical symptoms are consequently milder, and the disease may remain undiagnosed throughout life.[40]

Diagnosis

The initial diagnosis of DI rests on the combination of an inappropriately dilute urine in the context of hypernatremic dehydration. Thus, simple blood and urine chemistries easily establish the diagnosis, yet it is often delayed. When these infants present with a history of vomiting, physicians may be mistakenly reassured by the large urine output, considered as inconsistent with severe dehydration rather than recognizing it as a warning sign in the context of clinical dehydration. Once the diagnosis of DI has been established, the nephrogenic form can then be proven by the absence of a response in urinary concentration after administration of desamino-8-D-arginine vasopressin (DDAVP), a vasopressin analog with high specificity for AVPR2. DDAVP can be administered orally, nasally, or by intramuscular or intravenous injection. The most critical complication of the DDAVP test is the development of acute hyponatremia in those patients who respond to the drug yet keep on drinking. The risk for this is low, as most patients will simply stop drinking with normalization of serum osmolality, but patients with habitual polydipsia and infants hungry for milk are at risk. Careful observation with a limit in the volume of fluid intake equal to the volume of urine produced during the test is key to avoidance of this complication. Intravenous injection has the advantage of the shortest observation period (2 hours) combined with certainty of administration (which is variable via the oral or nasal route) and the ability to distinguish between X-linked and autosomal NDI. As patients with autosomal NDI still express AVPR2, they will show the vascular effects in the form of a small drop in BP, an increase in heart rate, and release of von Willebrand factor, whereas these changes are absent in patients with AVPR2 mutations. Serum levels of DDAVP with other modes of administration are not sufficient to appreciate these changes. A well-established protocol uses 0.3 μg/kg DDAVP intravenously, the same dose as used by hematologists in mild forms of hemophilia or von Willebrand disease.[39] A normal renal response to DDAVP is an increase in urine osmolality to greater than 800 mOsm/kg, whereas patients with NDI show no response and urine osmolality remains less than 200 mOsm/kg. An intermediate response may represent partial NDI or simply a washout of the medullary concentration gradient. The latter can be confirmed by achieving a urine osmolality of greater than 800 mOsm/kg after repeated administration of DDAVP. As normal urinary concentrating ability only develops during the first year of life, an intermediate response may be normal in infants.

Differential Diagnosis

Patients with central DI or habitual polydipsia can be distinguished by a normal response to DDAVP. Patients with secondary inherited NDI typically have other electrolyte abnormalities, such as hypokalemia and hypercalciuria, in addition to other features typical for the primary diagnosis, such as polyhydramnios in Bartter syndrome and hypertension in AME. Onset of polyuria later in childhood or adulthood argues for acquired forms of NDI. Patients with partial NDI represent a diagnostic challenge. Further blood tests and a renal ultrasound can help exclude underlying structural or parenchymal renal disease, such as nephronophthisis or tubulointerstitial or cystic kidney disease. Genetic testing can help confirm the diagnosis.

Treatment

Management of NDI concentrates on reduction of polyuria, avoidance of episodes of dehydration, and provision of sufficient calories to allow normal physical growth and development. A key factor in the determination of overall urine output is the osmotic load, which mainly derives from protein and salt intake. The higher the osmotic load, the more water is required to excrete it. Involvement of the dietician thus is key, especially in infancy, to minimize osmotic load, yet ensuring that protein and caloric intake meets the recommended daily intake.

A goal of 15 mOsm/kg is typically achievable; consequently, a patient with an average urine osmolality of 100 mOsm/kg would need 150 mL/kg/day to excrete this load.

Thiazide diuretics, such as hydrochlorothiazide (1–2 mg/kg/12 h), when combined with reduction of salt intake, are effective in reducing urine output. Thiazides inhibit salt reabsorption in DCTs, which leads to mild volume depletion. Hypovolemia stimulates fluid reabsorption in PTs, thereby diminishing water delivery to CDs. Combination therapy with amiloride (0.1–0.2 mg/kg every 8–12 hours) helps control the potential hypokalemia from thiazide treatment and may enhance the antipolyuric effect.

Prostaglandin synthase inhibitors are typically the most effective drugs in reducing urine volume, at least during infancy, probably by enhancing proximal sodium reabsorption. Indomethacin (1–2 mg/kg/day) is used most often but may reduce glomerular filtration rate and cause gastrointestinal side effects. Selective COX2 inhibitors, such as celecoxib, thus may be preferable, but their efficacy in NDI is so far anecdotal.

Nephrogenic Syndrome of Inappropriate Antidiuresis

NSIAD represents the mirror image of NDI and is the inherited equivalent to the syndrome of inappropriate antidiuresis (SIADH).

Pathogenesis

NSIAD is due to gain-of-function variants in *AVPR2,* leading to constitutive activation of the urinary concentrating mechanism in the CD, irrespective of serum osmolality.[41] The inheritance is X-linked dominant, although males are typically more severely affected.[42] Specific variants in genome-wide association studies, encoding the stimulatory G-protein alpha subunit, have also been associated with NSIAD and are not always associated with additional endocrine abnormalities.[43,44]

Clinical Features, Diagnosis, and Treatment

NSIAD is characterized by an inappropriately concentrated urine (>100 mOsm/kg) in the context of hyponatremia and hypoosmolality. In contrast to SIADH, vasopressin levels are adequately suppressed. Genetic testing will confirm the diagnosis. Severely affected patients (typically males) may have recurrent episodes of severe symptomatic hyponatremia, whereas others may have no apparent symptoms. Avoidance of fluid intake, mediated by low serum osmolality, constitutes simple and intuitive treatment and may explain the absence of symptoms in some patients.[42] Fluid restriction is difficult in infancy, as fluid and caloric intake is coupled. Increasing the osmotic load (e.g., by urea supplementation) may help symptomatic patients. AVPR2 antagonists (aquaretics) are typically not helpful, as there is no vasopressin excess.

SELF-ASSESSMENT QUESTIONS

1. Which of the following conditions is characteristically associated with hypocalciuria?
   A. Bartter syndrome
   B. Liddle syndrome
   C. Gitelman syndrome
   D. Gordon syndrome
2. The treatment of Bartter syndrome may include which of the following?
   A. Amiloride
   B. Prostaglandin synthesis inhibitors
   C. KCl supplementation
   D. NaCl supplementation
3. Hypokalemic, hypochloremic metabolic alkalosis is a sign of:
   A. aldosterone deficiency.
   B. enhanced Na transport through the epithelial sodium channel in the collecting duct.
   C. pseudohypoaldosteronism type 1.
   D. pseudohypoaldosteronism type 2.
4. Which of the following is the best suited maintenance fluid solution in patients with nephrogenic diabetes insipidus needing intravenous fluids?
   A. 0.9% saline
   B. Hartman solution
   C. 5% dextrose in water
   D. 0.45% saline

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